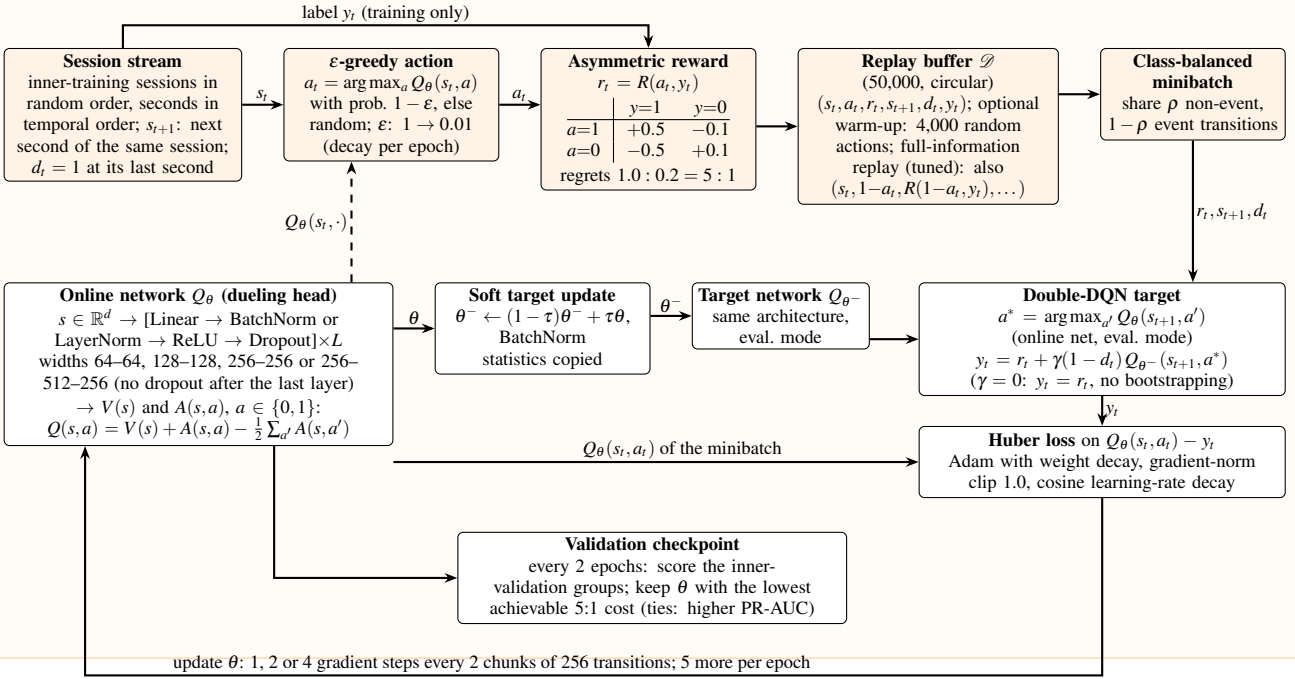

Training on the inner-training groups (labels used)
label y_t (training only)
Session stream
inner-training sessions in random order, seconds in temporal order; s_{t+1}: next second of the same session; $d_t = 1$ at its last second
s_t
ε-greedy action
$a_t = \arg\max_a Q_\theta(s_t, a)$ with prob. $1-\varepsilon$, else random; ε: $1 \to 0.01$ (decay per epoch)
a_t
Asymmetric reward
$r_t = R(a_t, y_t)$
y=1 y=0
a=1 +0.5 −0.1
a=0 −0.5 +0.1
regrets 1.0 : 0.2 = 5 : 1
Replay buffer $\mathcal{D}$
(50,000, circular)
$(s_t, a_t, r_t, s_{t+1}, d_t, y_t)$; optional warm-up: 4,000 random actions; full-information replay (tuned): also $(s_t, 1-a_t, R(1-a_t, y_t), \ldots)$
Class-balanced minibatch
share ρ non-event, $1-\rho$ event transitions
r_t, s_{t+1}, d_t
$Q_\theta(s_t, \cdot)$
Online network Q_θ (dueling head)
$s \in \mathbb{R}^d \to$ [Linear $\to$ BatchNorm or LayerNorm $\to$ ReLU $\to$ Dropout]$\times L$ widths 64–64, 128–128, 256–256 or 256–512–256 (no dropout after the last layer) $\to V(s)$ and $A(s,a)$, $a \in \{0,1\}$: $Q(s,a) = V(s) + A(s,a) - \frac{1}{2}\sum_{a'} A(s,a')$
θ
Soft target update
$\theta^- \leftarrow (1-\tau)\theta^- + \tau\theta$, BatchNorm statistics copied
θ^-
Target network Q_{θ^-}
same architecture, eval. mode
Double-DQN target
$a^* = \arg\max_{a'} Q_\theta(s_{t+1}, a')$ (online net, eval. mode)
$y_t = r_t + \gamma(1-d_t)\, Q_{\theta^-}(s_{t+1}, a^*)$
($\gamma = 0$: $y_t = r_t$, no bootstrapping)
y_t
$Q_\theta(s_t, a_t)$ of the minibatch
Huber loss on $Q_\theta(s_t, a_t) - y_t$
Adam with weight decay, gradient-norm clip 1.0, cosine learning-rate decay
Validation checkpoint
every 2 epochs: score the inner-validation groups; keep θ with the lowest achievable 5:1 cost (ties: higher PR-AUC)
update θ: 1, 2 or 4 gradient steps every 2 chunks of 256 transitions; 5 more per epoch
Inference on validation and test groups (no labels, no updates)
State s_t: causal features of seconds 0..t (frozen featurizer and scaler)
Checkpointed Q_θ of each of the 3 inner models
Score $\hat{p} = \frac{\Delta Q + 0.2}{1.2}$, $\Delta Q = Q(s,1) - Q(s,0)$; mean over the 3 models
Threshold t_{D1} chosen on pooled inner-OOF scores
Alarm if $\hat{p} \geq t_{D1}$